



MANIPAL SCHOOL OF INFORMATION SCIENCES

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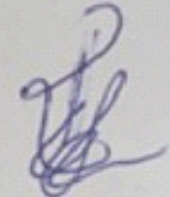
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Master of Engineering - M.E in Computer Science and Engineering (Embedded Systems)

Course Plan

Course Name	:	Microcontrollers and its Applications
Course Code	:	ESE 5103
Academic Year	:	2026-2027
Semester	:	I
Name of the Course Coordinator	:	Ms. Ravikala Kamath
Name of the Program Coordinator	:	Dr. B Dinesh Rao

	
Date:	1.8.2026
Signature of Program Coordinator	
with Date	

	
Date:	30.7.2026
Signature of Course Coordinator	
with Date	



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Program Education Objectives (PEOs)

The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering - M.E in Computer Science and Engineering (Embedded Systems), program are as follows

PEO No.	Education Objective
PEO 1	Enable to draw upon fundamental and advanced knowledge to apply analytical and computational approaches to solve technological problems in embedded systems.
PEO 2	Introduce state of art technologies in the area of embedded systems and inculcate ethical practices to make industry-ready professionals.
PEO 3	Promote scientific and societal advancement through research and entrepreneurship.



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Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering - M.E in Computer Science and Engineering (Embedded Systems), graduates will be able to:

PO 1	An ability to independently carry out research /investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
PO 4	An ability to develop and implement embedded systems requirements based on theoretical principles and practical knowledge.
PO 5	An ability to demonstrate knowledge of the underlying principles and evaluation methods for analyzing and decision-making.



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1. Course Plan

1.1 Primary Information

Course Name	:	Microcontrollers and its Applications [ESE 5103]
L-T-P-C	:	3-0-0-3
Contact Hours	:	36 Hours
Pre-requisite	:	Basic knowledge of controller and c programming
Core/ PE/OE	:	Core

1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

CO	At the end of this course, the student should be able to:	No.ofContact Hours	Program Outcomes (POs)	BL
CO1	Apply the knowledge of Microcontrollers to build embedded systems.	16	PO3	3
CO2	Apply the concepts of timers, counters, ADC and DAC for interfacing.	10	PO3	3
CO3	Analyse the features of SPI, I2C, and parallel buses and interface sensors and actuators.	10	PO4	4



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1.3 Assessment Plan

Components	Mid semester exam (MSE)	Flexible assessments	End semester exam (ESE) / Makeup exam
Duration	90 minutes	2	180 minutes
Weightage	0.3	0.2	0.5
Typology of questions	Apply	Apply, Analyze	Apply, analyze
Pattern	Each question carry 3 /4 /5 / 10 marks	Each question carry 3 /4 /5 / 10 mark covers co1, co2,co3	Each question carry 3 /4 /5 / 10 marks
Schedule	As per academic calendar.	Assignment submission: september 30th, october 25th	As per academic calendar.
Topics covered	Component to basic programming of ports, timers, modes , Arm, timers, controllers	Component to basic programming of ports, timers, ADC, DAC, UART, SPI, I2C ALL	Comprehensive examination covering the full syllabus. Students are expected to answer all questions.



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1.4 Lesson Plan

L. No.	TOPICS	Course Outcome Addressed
L0	Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books.	---
L1	Introduction to Microprocessor & Microcontrollers: Comparison, Variants, Types, General, ASIC, PLD	C01
L2	Introduction to Motherboard (Desktop), Introduction to Embedded Board, Compare and Contrast	C01
L3	Application Types, Single Tasking, Multitasking, Multi-Application	C01
L4	Introduction to ARM Microcontrollers: Programming Model,	C01
L5	Processor Modes, ARM vs Thumb,	C01
L6	Introduction to LPCxxxx Microcontrollers	C01
L7	Features, Detailing of Pins, Memory Map Concepts	C01
L8	RAM & ROM - Interrupts Concepts (Internal & External).	C01
L9	Reset Circuitry – Crystals: Introduction to GPIO,	C01
L10	Registers, Input /Output Configuration,	C01
L11	Pull Up and Pull-Down Resistor Concept,	C01



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L12	Interfacing with LED, Interfacing Push Buttons	C01
L13	Working of LCD,	C01
L14	tepper Motor	C01
L15	DC Motor	C01
L16	Types of Relays	C01
L17	Interfacing	C01
L18	Introdction to Timer, understanding the working	C02
L19	functionalits of timers	C02
L20	programming aspects of timer	C02
L21	Counter Introduction, Configuration,	C02
L22	Programming.	C02
L23	Introduction to ADC	C02
L24	Introduction to DAC	C02
L25	Programming	C02



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L26	MSE	C01,C02
L27	Types,Chips	C02
L28	Register Configuration, Interfacing	C02
L29	Serial vs Parallel Bus-ompare and Contrast – Terminology:	C03
L30	Baud Rate, Bit Rate, RS232, DB9 handshaking concepts,	C03
L31	Configuring Registers,	C03
L32	Programming for UART module.	C03
L33	Programming for UART module.	C03
L34	Introduction to SPI and I2C Protocol	C03
L35	Detailed Discussion, Bit Banging, Interfacing with SPI	C03
L36	I2C Devices, RTC	C03
L37	INTERFACING ADC /DAC.	C03
L38	INTERFACING ADC /DAC.	C03



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1.5 References

1. William Hohl, Christopher Hinds, "ARM Assembly Language: Fundamentals and Techniques", 2nd Edition, ISBN-13: 978-1482229851, ISBN-10: 1482229854.
2. Andrew Sloss, Dominic Symes, Chris Wright, "ARM System Developer's Guide: Designing and Optimizing System Software", 1st Edition, The Morgan Kaufmann Series in Computer Architecture and Design, ISBN-13: 978-1558608740, ISBN-10: 1558608745.
3. David Seal, "ARM Architecture Reference Manual", 2nd Edition, Addison-Wesley Professional.
4. Steve Furber, "ARM System-on-Chip Architecture", 2nd Edition, Addison-Wesley Professional, ISBN-13: 078-5342675191, ISBN-10: 0201675196.
5. Douglas V. Hall, "Microprocessors and Interfacing", McGraw Hill Education, ISBN-10 1259006158, ISBN-13 9781259006159, 2012.
6. Websites & Transaction Papers.
7. MOOC: <https://www.coursera.org/learn/armv8-m-architecture-fundamentals#syllabus>

1.6 Other Resources (Online, Text, Multimedia, etc.)

1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.



1.7 Course Timetable

I Semester						Room: LH-01, LH-08			
	8-9	9-10	10-11	11-12	12-1	1-2	2-3	3-4	4-5
Monday								MCA	
Tuesday									
Wednesday								MCA	
Thursday								MCA	
Friday									
Saturday									



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1.8 Assessment Plan

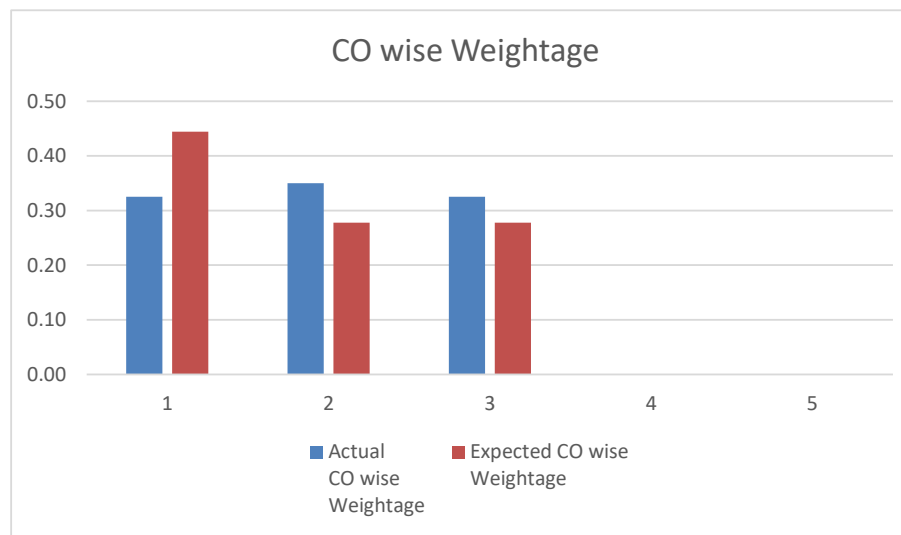
COs		Marks & Weightage				
CO No.	CO Name	MSE Marks (Max. 50)	Assignment Marks (Max. 20)	ESE Marks (Max. 100)	Actual CO wise Weightage	Expected CO wise Weightage
C01	Apply the knowledge of Microcontrollers to build embedded systems.	25	5	25	0.33	0.44
C02	Apply the concepts of timers, counters, ADC and DAC for interfacing.	25	5	30	0.35	0.28
C03	Analyse the features of SPI, I2C, and parallel buses and interface sensors and actuators.	0	10	45	0.33	0.28
	Marks (weightage)	0.3	0.2	0.5	1.00	1.00



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Note:

- In-semester Assessment is considered as the Internal Assessment (IA) in this course for 50 marks, which includes the performances in class participation, assignment work, class tests, MSE, quizzes etc.
- ESE for this course is conducted for a maximum of 100 and the same will be scaled down to 50 for grading.
- ESE marks for a maximum of 50 and IA marks for a maximum of 50 are added for a maximum of 100 marks to decide upon the grade in this course.

$$\text{Weightage for CO1} = (\text{MSE marks for CO1} / 1.6666 + \text{Assignment marks for CO1} / 1.0 + \text{ESE marks for CO1} / 2) / 100$$



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1.9 Assessment Details

The assessment tools to be used for the Current Academic Year (CAY) are as follows:

Sl. No.	Tools	Weightage	Frequency	Details of Measurement (Weightage/Rubrics/Duration, etc.)
1	MSE	0.3	1	- Performance is measured using MSE attainment level. - Reference: question paper and answer scheme. - MSE is assessed for a maximum of 50 marks and scaled down to 30 marks.
2	Assignments	0.2	2	- Performance is measured using assignments/quiz attainment level. - Assignments/quiz are evaluated for a maximum of 20 marks.
3	ESE	0.5	1	- Performance is measured using ESE attainment level. - Reference: question paper and answer scheme. - ESE is assessed for a maximum of 100 marks and scaled down to 50 marks.

1.10 Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5
CO1			Y		
CO2			Y		
CO3				Y	
Average Articulation Level			*	*	